AI-Optimized IDDQ Test Time Reduction

Technical document

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1. Executive summary

Machine learning can cut IDDQ test time by removing redundant vectors, shortening settling waits and skipping low-risk die, while keeping or improving defect screening. IDDQ remains one of the slowest tests per vector because each quiescent-current measurement must wait for the supply to settle. In advanced nodes, high and variable background leakage also makes fixed pass/fail limits weak.

This document describes six complementary optimization levers:

1. **Vector reduction** — keep only the IDDQ states that add unique defect detection.
2. **Settling-time prediction** — estimate the settled current from early transient samples.
3. **Adaptive per-die test** — run a reduced IDDQ set on die predicted to be low-risk.
4. **ML-based outlier screening** — replace fixed limits with leakage-normalized, multivariate screens.
5. **Spatial prediction** — model wafer leakage maps so only deviations need full measurement.
6. **Flow ordering** — place high-fallout vectors early so failing die stop sooner.

The main risks are test escapes (higher DPPM), model drift across lots and process changes, and customer or automotive acceptance. Every lever therefore ships with guard-bands, a full-test audit sample and downstream escape tracking.

All time and savings figures in this document are illustrative planning values. They must be replaced with measured data from your product and tester before any decision.

2. Background: IDDQ testing

2.1 Principle

A fault-free static CMOS circuit draws only leakage current from VDD when no inputs are switching. Many physical defects create a conducting path from VDD to ground in a given logic state, which raises the quiescent supply current (IDDQ) by orders of magnitude above leakage in older nodes. IDDQ testing applies a set of vectors that put the circuit into chosen quiescent states, waits for transients to decay, then measures supply current per state.

2.2 Defects IDDQ detects well

- Bridging faults (resistive shorts between nets), including those that do not cause a logic error.
- Gate-oxide shorts and pinholes.
- Transistor stuck-on and leaky junction defects.
- Floating gates and some open defects that leave a node at an intermediate voltage.
- Latent reliability defects that may escape functional and scan tests but fail early in the field.

2.3 Why IDDQ is harder in advanced nodes

- **Background leakage is high.** Subthreshold, gate and junction leakage grow as dimensions shrink, so the defect current is a small fraction of total current.
- **Leakage varies widely.** Die-to-die and wafer-to-wafer process variation spreads normal IDDQ over a wide range, so a single fixed limit either fails good die or passes defective ones.
- **Strong temperature and voltage dependence.** Leakage changes sharply with temperature and VDD, so test conditions must be tightly controlled.
- **Many power domains.** Modern SoCs have multiple rails and power-gated islands, multiplying measurements.

2.4 Derived IDDQ metrics

To separate defects from normal leakage, production flows use relative metrics rather than raw current:

- **Delta-IDDQ** — the difference between IDDQ of consecutive or sorted vectors. A defect active only in some states creates a step.
- **Current ratio** — max IDDQ divided by min IDDQ for a die. Normal die have a stable ratio regardless of absolute leakage.
- **Nearest-neighbor residual (NNR)** — the die's IDDQ minus an estimate from surrounding die on the wafer, removing spatial leakage trends.
- **Multi-condition IDDQ** — IDDQ at several voltages or temperatures, where defects scale differently from leakage.

3. Test time breakdown and cost model

IDDQ time per die is dominated by settling wait multiplied by the number of vectors and rails. The total can be modeled as:

$$T_{IDDQ} = \sum_{r=1}^{R} \sum_{v=1}^{N_v} \left(t_{apply} + t_{settle,r,v} + t_{meas,r} \right) + t_{range} + t_{retest}$$

where R is the number of measured rails, N_v the number of IDDQ vectors, t_{apply} the scan load or pattern application time, t_{settle} the wait for current to stabilize, t_{meas} the measurement integration time, t_{range} the overhead of measurement-unit range changes, and t_{retest} the expected time spent re-measuring marginal die.

3.1 Contributors

Contributor	Typical driver	AI lever that addresses it
Settling time per vector	Decoupling capacitance, tester PMU response, power-gating wake-up	Settling-time prediction (4.2)
Number of vectors	Fault coverage targets, legacy test programs	Vector reduction (4.1), adaptive test (4.3)
Number of rails and conditions	Power domains, multi-VDD or multi-temperature screens	Adaptive test (4.3), feature ranking (6)
Retests of marginal die	Fixed limits close to the leakage distribution	Outlier screening (4.4)
Average time to first fail	Vector order	Flow ordering (4.6)
Die measured at full depth	Every die tested the same way	Spatial prediction (4.5), adaptive test (4.3)

3.2 Savings model

Expected saving per die combines the levers multiplicatively on the parts each one touches:

$$T_{new} = R \cdot N'_v \cdot \left(t_{apply} + \alpha\, t_{settle} + t_{meas} \right) \cdot \left(p_{full} + (1 - p_{full})\, \beta \right)$$

where N'_v is the reduced vector count, α the fraction of settling time still waited after prediction (0 to 1), p_{full} the fraction of die that still receive the full set, and β the relative length of the reduced set for low-risk die.

The cost value of the saving is:

$$\Delta C = \left(T_{IDDQ} - T_{new}\right) \times \text{tester cost per second} \times \text{annual volume} - C_{escape}$$

C_{escape} is the expected cost of additional test escapes (returns, field failures, customer penalties). It must be estimated conservatively and drives the guard-band choices in Section 7.

Open question: current per-die IDDQ time, vector count, rail count and tester cost per second for the target product are needed to turn this model into a business case.

4. AI/ML optimization techniques

The six levers below are independent and can be deployed in stages; vector reduction and settling prediction are offline or low-risk and usually come first.

4.1 IDDQ vector selection and reduction

Goal: the smallest vector set that keeps defect detection.

Inputs

- Fault-simulation coverage per vector for bridging, pseudo-stuck-at and leakage fault models.
- Historical per-vector IDDQ for passing and failing die across many lots.
- Unique-fail attribution: which die failed only on a given vector.

Methods

- *Greedy set cover* — repeatedly pick the vector that adds the most uncovered faults or historical fails. Fast and explainable baseline.
- *Genetic algorithms or simulated annealing* — search vector subsets with a fitness function of coverage minus a time penalty.
- *Reinforcement learning* — an agent adds or removes vectors, rewarded for retained coverage and reduced time. Useful when vectors differ in settling time or rail count.
- *Correlation clustering* — cluster vectors whose IDDQ readings are highly correlated across die; keep one representative per cluster.

Output: a ranked vector list and a coverage-versus-count curve, from which the team picks a knee point with margin.

4.2 Settling-time prediction

Goal: stop waiting before full settling by predicting the final current.

Approach

1. Capture several fast samples of supply current during the early transient after each vector.
2. Fit a model that maps the early samples to the fully settled value measured in characterization.
3. In production, accept the prediction when its confidence interval is narrow and far from the limit; otherwise wait for full settling.

Model options

- Physics-informed fit: sum of one or two decaying exponentials plus a constant, solved for the constant.
- Gradient-boosted trees or a small neural network on sample values, slopes, vector ID, rail and temperature.
- Quantile regression or conformal prediction to produce calibrated confidence bounds.

Constraints: requires tester hardware that can sample the transient quickly, and characterization data with full settling for training.

4.3 Adaptive per-die IDDQ

Goal: run the full IDDQ set only where it is likely to matter.

Inputs available before IDDQ

- Parametric tests: leakage at other conditions, Vmin, Fmax, ring-oscillator frequencies, process monitor readings.
- Wafer position (radius, edge distance, reticle position) and neighbor results.
- Lot and wafer context: equipment history, inline metrology if available.

Method

- A classifier or anomaly score estimates each die's risk of an IDDQ fail.
- Die below a risk threshold run a reduced vector set; die above it, or with missing inputs, run the full set.
- A fixed random fraction of die always run the full set as an audit sample.

Infrastructure: real-time adaptive test hooks on the tester or test-cell controller that can read upstream results and switch flows within the die's test time.

4.4 ML-based outlier screening

Goal: catch more true defects with fewer measurements by screening against what normal looks like for that die.

Methods

- *Statistical baselines:* delta-IDDQ, current ratio and NNR (Section 2.4).
- *Multivariate distance:* Mahalanobis distance across all vectors, rails and conditions, using a robust covariance estimate.
- *Regression residuals:* predict expected IDDQ from leakage-correlated parametrics; screen on the residual.
- *Unsupervised anomaly detection:* isolation forest, one-class SVM or autoencoder reconstruction error on IDDQ signatures.
- *Gaussian process or spatial models:* expected IDDQ from wafer position, screening the residual.

Benefit to test time: better separation lets the program drop redundant conditions and reduces marginal retests, because fewer die land near a fixed limit.

4.5 Spatial and wafer-level prediction

Goal: reduce the number of die that need full-depth measurement.

- Measure the full IDDQ set on a sample of die per wafer.
- Reconstruct the wafer leakage map using Gaussian process regression or compressed-sensing (virtual probe) methods.
- Measure the remaining die with a short set and compare against their predicted value; any deviation triggers the full set.

4.6 Test-flow ordering

Goal: lower average time per die by failing defective die sooner.

- Rank vectors by historical fail probability per unit time.
- Place the highest-yield-loss vectors first, subject to dependencies.
- Combine with stop-on-first-fail where datalogging requirements allow.

5. System architecture and data pipeline

The system splits into an offline loop that learns from full-test data and an online loop that decides each die's flow within its test time.

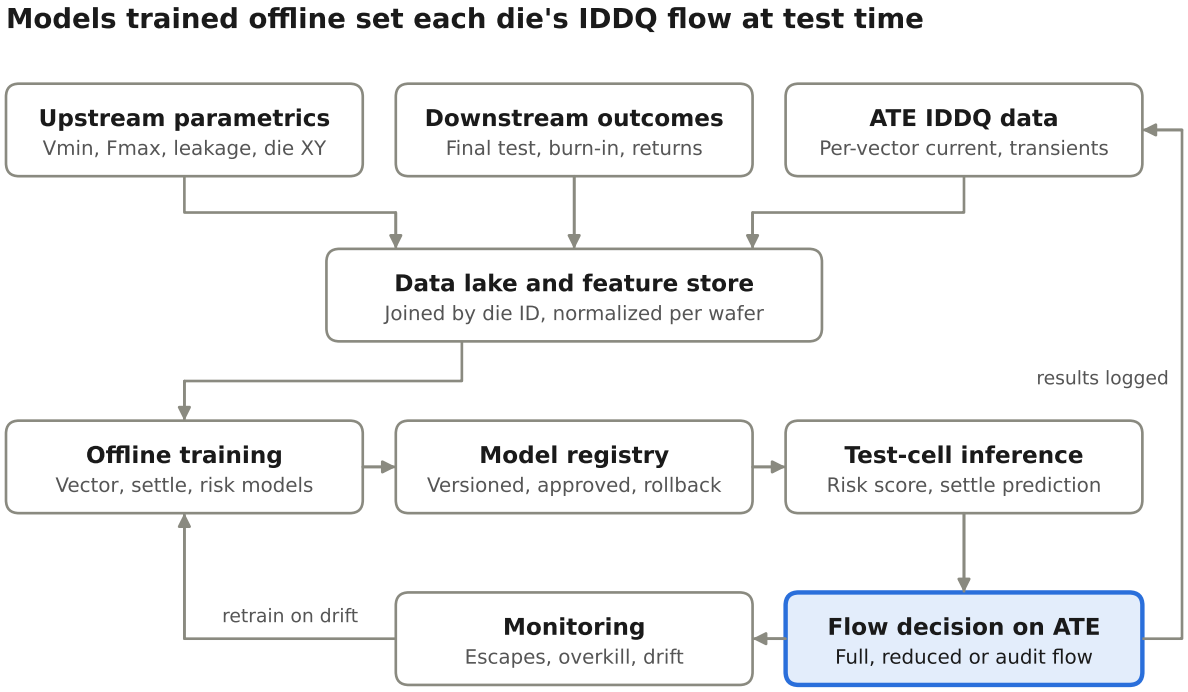


Figure 1: System architecture: data, training, deployment and monitoring

Every die's measured results return to the data lake, so the audit sample and downstream outcomes keep retraining and monitoring current.

5.1 Components

Component	Responsibility	Latency need
Data lake and feature store	Join ATE, parametric and outcome data by die ID; compute features	Hours (batch)
Offline training	Fit vector-selection, settling and risk models; tune thresholds	Days
Model registry	Store approved model versions, thresholds and test-program links	n/a
Test-cell inference	Score each die and predict settled current on the tester or cell controller	Milliseconds per die
Flow decision	Choose full, reduced or audit flow; log the decision and model version	Milliseconds
Monitoring	Track escapes, overkill, drift and time saved; trigger fallback	Near real time

5.2 Integration notes

- Inference must fit inside existing test time; lightweight tree models or lookup tables are preferred at the edge.
- Upstream data for adaptive decisions must be available when the die reaches IDDQ (same insertion, or fed forward from wafer sort).
- The test program needs a flow-switch hook and must log the decision per die for traceability.
- Data formats such as STDF should carry die ID, model version and flow chosen.

6. Model development

Models are trained offline on full-test data and judged first on escapes (missed defective die), then on time saved.

6.1 Data requirements

- Full IDDQ data for every vector, rail and condition, without stop-on-fail, over many lots and wafers.
- Settling transients sampled at high rate for a characterization subset (for 4.2).
- Upstream parametric results and die coordinates linked by unique die ID (ECID or lot/wafer/x/y).
- Downstream outcomes linked to the same die: final test, burn-in, system-level test and customer returns. These are the labels that define a true defect.
- Tester, site, probe card and temperature metadata, to detect equipment effects.

6.2 Feature engineering

Feature group	Examples
Raw IDDQ	Per-vector, per-rail current; min, max, median per die
Relative IDDQ	Delta-IDDQ steps, max/min ratio, sorted-vector curve shape
Spatial	Radius, edge distance, reticle position, NNR, neighbor fail counts
Process proxies	Ring-oscillator frequency, leakage monitors, Vmin, Fmax
Condition response	IDDQ slope versus VDD or temperature
Transient (4.2 only)	Early sample values, slopes, fitted time constants
Context	Tester, site, lot, wafer, test time of day

Normalize leakage-sensitive features per wafer or per lot so models learn deviations rather than process shifts.

6.3 Algorithm choice

Task	Recommended starting point	Reason
Vector selection	Greedy set cover, then GA refinement	Explainable, fast, easy to audit
Settling prediction	Exponential fit plus quantile gradient boosting	Physics prior with calibrated bounds
Per-die risk	Gradient-boosted trees (XGBoost or LightGBM)	Strong on tabular data, feature importance available
Outlier screening	Robust Mahalanobis, isolation forest	Works with few labeled defects
Spatial maps	Gaussian process regression	Uncertainty estimates per die

Defective die are rare, so class imbalance must be handled with resampling, class weights or anomaly-detection framing.

6.4 Training and validation protocol

1. Split by lot and by time, never by random die, so validation reflects future production.
2. Hold out the most recent lots as a final test set.

3. Tune thresholds for zero or near-zero escapes on the validation set, then report time saved at that threshold.
4. Test on lots from different fab conditions, testers and temperatures to check robustness.
5. Run in shadow mode in production: the model predicts but the full test still runs, so escapes can be counted directly.

6.5 Validation metrics

Metric	Definition	Target direction
Escape rate	Defective die the reduced flow would pass, per million	As low as baseline or lower
Overkill	Good die the new screen would fail	No increase over baseline
Recall at threshold	Share of known IDDQ fails caught	Close to 100%
Test time per die	Mean and 95th percentile seconds	Lower
Prediction error (4.2)	Predicted vs settled IDDQ, absolute and relative	Within measurement repeatability
Calibration	Observed vs predicted fail rate per risk bin	Diagonal
Drift indicators	Feature distribution shift per lot (e.g. PSI)	Below alert limit

7. Quality, risk and compliance

No test-time reduction is released unless the escape rate stays at or below the current flow's, proven in shadow mode and audited continuously in production.

7.1 Risk register

Risk	Effect	Mitigation
Test escapes	Higher DPPM, field failures, returns	Zero-escape threshold tuning, audit sample, downstream escape tracking
Model drift	Predictions degrade after process, mask or equipment change	Drift monitors per lot, automatic fallback to full flow, scheduled retraining
Missing or bad upstream data	Wrong risk score for a die	Missing data always triggers the full flow
Tester or site effects	One site biases readings	Site-level normalization, site in features, per-site monitoring
Reliability defects with small IDDQ signature	Latent fails escape reduced screen	Keep reliability-critical vectors and conditions fixed; correlate with burn-in data
Black-box rejection by customers	Program not approved	Explainable models, documented rules, feature importance reports
Overkill	Yield loss from aggressive outlier screens	Track overkill as a release metric; review rejected die

7.2 Guard-banding rules

- A prediction is used only when its confidence bound sits clear of the limit by a set margin.
- Die with any missing input, unusual context or new equipment run the full flow.

- A fixed audit fraction of die always runs the full flow, chosen randomly so it cannot be gamed by the model.
- Any lot that trips a drift alert reverts to the full flow until reviewed.

7.3 Automotive and high-reliability products

Automotive products qualified under AEC-Q100 and used in ISO 26262 systems typically have strict DPPM targets and change-control requirements. For these products:

- Treat any test reduction as a formal process change requiring customer notification or approval.
- Prefer outlier screening improvements (which tighten screening) over test skipping.
- Keep full traceability: model version, threshold and flow decision recorded per die.
- Validate against burn-in and field data over a longer period before release.

7.4 Monitoring in production

- Daily dashboards: test time saved, audit-sample escape rate, overkill, drift indicators.
- Escape investigation loop: every downstream fail of a die that took the reduced flow is analyzed and fed back to training.
- Model governance: version control, approval records and rollback plan for every deployed model.

8. Implementation roadmap and KPIs

Rollout runs in four phases, and each gate must pass on measured data before the next phase starts.

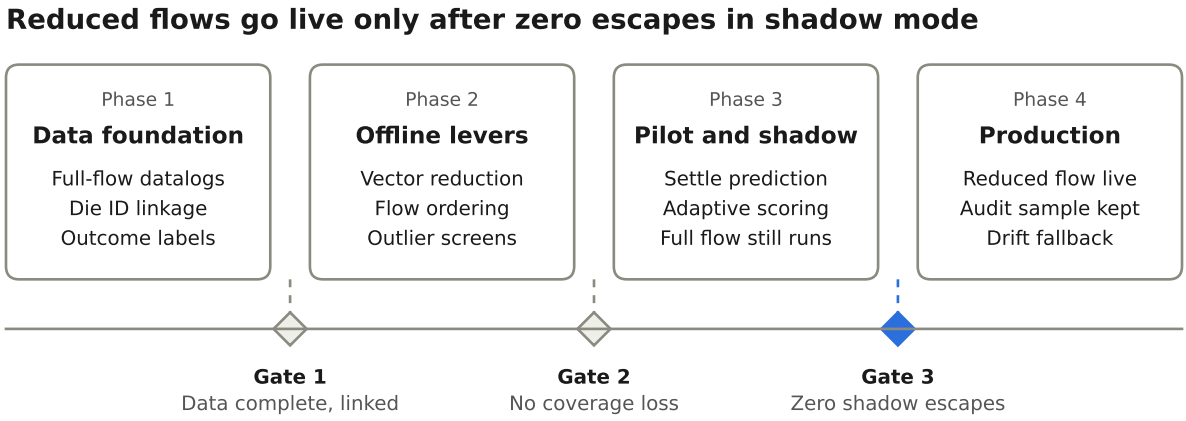


Figure 2: Implementation roadmap: 4 phases, 3 gates (not to scale)

Phase durations depend on lot volume and how quickly downstream outcome data accumulates; plan them once Phase 1 shows how fast labeled data arrives.

8.1 Gate criteria

Gate	Pass criteria
Gate 1	Full IDDQ datalogs for enough lots; die ID linked to upstream and downstream data; data quality checks pass
Gate 2	Reduced vector set and new screens show no loss of detected fails on held-out lots; overkill not increased

Gate	Pass criteria
Gate 3	Shadow mode over agreed lot count shows zero additional escapes; settling predictions within measurement repeatability; customer approval where required

8.2 KPIs

KPI	Measure	Reported
IDDQ test time per die	Mean and 95th percentile, seconds	Per lot
Total test time per die	Seconds, all insertions	Per lot
Escape rate	DPPM from audit sample and downstream test	Weekly
Overkill	Good die failed by new screens	Weekly
Reduced-flow share	Share of die taking the reduced flow	Per lot
Fallback rate	Share of die or lots reverted to full flow	Per lot
Drift alerts	Count and lots affected	Daily
Cost saved	Tester seconds saved times cost per second	Monthly

9. Glossary and further reading

9.1 Glossary

Term	Meaning
IDDQ	Quiescent supply current of a CMOS circuit when inputs are static
Delta-IDDQ	Difference in IDDQ between vectors, used to remove background leakage
NNR	Nearest-neighbor residual: die IDDQ minus a neighbor-based estimate
DPPM	Defective parts per million shipped
Escape	A defective die that passes test
Overkill	A good die that fails test
ATE	Automatic test equipment
PMU	Parametric measurement unit used to measure supply current
BICS	Built-in current sensor, an on-chip IDDQ monitor
ECID	Electronic chip ID, a unique die identifier
Adaptive test	Changing a die’s test flow in real time based on its own or neighbors’ data
PSI	Population stability index, a drift metric
Shadow mode	Running a model alongside the full flow without acting on its decisions

9.2 Further reading

These are well-known works in the field, listed from memory; please verify titles and details before citing.

- R. Rajsuman, “Iddq testing for CMOS VLSI,” *Proceedings of the IEEE*, 2000.
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